BASAVESHWAR VEERASHAIVA VIDYAVARDHAKA SANGHA, BAGALKOT

AMRUTA INSTITUTE OF ENGINEERING, MANAGEMENT SCIENCES & POLYTECHNIC

(Approved by AICTE, New Delhi, Recognized by Government of Karnataka and Department of Technical Education)

AIeMS innovation Integrity Excellence

INS - FORMAT - 10

Lesson Plan: year 2025-2026

Institution Name: AIEMS POLYTECHNIC	Institution code: 592	
With Effect from: JUNE 2025	Name of the Program: Computer Science & Engineering	Semester: III
Academic year:2025-26	Name of the Course Coordinator: Amulya K S	Course Code: 20CS34P

Week	Session No	CO	PO		Concept Planned	Planned date	Hours allotted	Concept covered	Teacher Activity	Student Activity	Pedagogy Adopted	Teaching Aids Used	Learning Outcomes Achieved	Remarks by Programme Coordinator
1	1	1	1, 4	L-1	Overview of DB: why a database? Purpose of database; Classification; Application; DBMS: features, providers;	30-06-2025	1	As per Planned	Lecture, Provide an exposure regarding Different platforms to collect information	Collecting the information about DBMS and Software installation	Lecturing, Demonstration, Activity, Interactive and Participative approach	Computers, Projector, Chalk & Talk, Youtube Videos	Students are familier with Overview of DB, purpose of database; Classification; Application; DBMS: features, providers	
	2			L-2	Functional components of DBMS; Types of DBMS architecture;View of data in DBMS; Database users; Role and responsibilities of DBA;	30-06-2025	1							
	3			L-3	Case study: Example of any database application, recruitment database	01-07-2025	1							
	4			T-1	1. Identify the drawbacks of file system and how DB enables us to overcome them. Identify distinguishable features of each of DBMS available in the market.	01-07-2025	1							

2	5	1	1, 3, 4	L-1	Data model; types; importance of data modeling; Overview of database design; phases of database design; database development life cycle;	07-07-2025	1	As per Planned	Split the class into groups of three and assign a task. Instruct students to portray the task given.	Designing different DBs group discussion	Lecturing, Demonstration, Activity, Interactive and Participative approach	Computers, Projector, Chalk & Talk, Youtube Videos	Students are familier with designing the database and ER model	
	6			L-2	Conceptual design: ER-Model: entity: types; attribute: types; relationships: types, constraints, Symbols and Notations;	07-07-2025	1							
	7			L-3	Case study: conceptual design for a set of specifications i. Restaurant ii. Retail shop iii. Recruitment iv. College v.Library	08-07-2025	1							
	8			T-1	Transform given n-ary relationship to binary relationship 2. Document the steps to create ER diagram. 3. Identify the components of ER model in the given requirements.	08-07-2025	1							
3	9	2	1, 3	L-1	Relational model: Overview; characteristics; Constraints: types; Operations;	14-07-2025	1	As per Planned	Split the class into groups of three and assign a task. Instruct students to portray the task given.	Group discussion, solving the given activities and collecting the information about given topic	Lecturing, Demonstration, Activity, Interactive and Participative approach	Computers, Projector, Chalk & Talk, Youtube Videos	Students are familier with designing the database and ER model,codd's rule and constraints	
	10			L-2	Advantages and Disadvantages; applications;	14-07-2025	1							
	11			L-3	Design anomalies; Features of good DB design	15-07-2025	1							
	12			T-1	Document the steps to create logical design Discuss and document Codd's 12 rules	15-07-2025	1							

4	13	2	1, 3	L-1	Functional dependency: overview, rules, types; Normalization: normalization process;	21-07-2025	1	As per Planned	Split the class into groups of three and assign a task. Instruct students to portray the task given.	Group discussion, solving the given activities and collecting the information about given topic	Lecturing, Demonstration, Activity, Interactive and Participative approach	Computers, Projector, Chalk & Talk, Youtube Videos	Students are familier with functional dependancies and Normalization	
	14			L-2	Importance of normalization;1NF,2NF,3NF	21-07-2025	1							
	15			L-3	Sufficient examples to understand the concept	22-07-2025	1							
	16			T-1	Explore and document other normal forms	22-07-2025	1							
5	17	3	1, 4	L-1	Database languages: types, commands/tasks in each type; Integrity constraints;	28-07-2025	1	As per Planned	Discussing constraints on different entities and Validate the above design against integrity constraints	Group discussion, solving the given activities and collecting the information about given topic and class test	Lecturing, Demonstration, Activity, Interactive and Participative approach	Computers, Projector, Chalk & Talk, Youtube Videos	Students are familier with different functions of DB and SQL commands and queries	
	18			L-2	MySQL/PostgreSQL: overview; features;	28-07-2025	1							
	19			L-3	Datatypes; Standardization guidelines	29-07-2025	1							
	20			T-1	Study and present the working of SQL optimizer Learn and report optimization techniques	29-07-2025	1							

6	21	3	1, 3, 4	L-1	Defining Data: DDL CREATE, ALTER, DROP different DB objects;	11-08-2025	1	As per Planned	1.Make groups of students 2.Assign different tasks regarding case study, Discuss the various databases	Solving different queries on different DBS functions,group discussion, activities and class test	Lecturing, Demonstration, Activity, Interactive and Participative approach	Computers, Projector, Chalk & Talk, Youtube Videos	Students are familier with different functions of DB and SQL commands	
	22			L-2	Temporary tables: types, create and use;	11-08-2025	1							
	23			L-3	External tables; Managing constraints	12-08-2025	1							
	24			T-1	Learn and demonstrate use of DISTINCT, ALL, IS NULL; Learn and present RIGHT JOIN;	12-08-2025	1							
7	25	3, 4	1, 3, 4	L-1	Insert, delete and update data Modifying data: UPDATE and DELETE	18-08-2025	1	As per Planned	Performing different operations on different databases	Solving different queries on different DBS functions,group discussion, activities and class test	Lecturing, Demonstration, Activity, Interactive and Participative approach	Computers, Projector, Chalk & Talk, Youtube Videos	Students are familier with different functions of DB and SQL commands , queries and transaction operations	
	26			L-2	Update anomalies; impact of constraints	18-08-2025	1							
	27			L-3	Querying of available data: SELECT; Aliases; sorting data: ORDER BY	19-08-2025	1							
	28			T-1	Identify the advantages of Cascading Referential Integrity Constraints	25-08-2025	1							

8	29	4	1, 4	L-1	Filtering data: WHERE, AND, OR	25-08-2025	1	As per Planned	Performing different operations on different databases	Solving different queries on different DBS functions,group discussion, activities	Lecturing, Demonstration, Activity, Interactive and Participative approach	Computers, Projector, Chalk & Talk, Youtube Videos	Students are familier with different functions of DB and SQL commands, queries and transaction operations	
	30			L-2	Row limiting clause IN, BETWEEN, LIKE;	26-08-2025	1							
	31			L-3	Joining table: INNER JOIN, LEFT JOIN	26-08-2025	1							
	32			T-1	Identify need of subqueries	28-08-2025	1							
9	33	4	1, 4	L-1	Grouping Data: Aggregate functions	28-08-2025	1	As per Planned	Performing different operations on different databases	Solving different queries on different DBS functions,group discussion, activities and class test	Lecturing, Demonstration,activities,Interactive and Participative approach	Computers, Projector, Chalk & Talk, Youtube Videos	Students are familier with different functions of DB and SQL commands , queries and transaction operations	
	34			L-2	GROUP BY, HAVING;	29-08-2025	1							
	35			L-3	Set operators: UNION, INTERSECT, MINUS;	29-08-2025	1							
	36			T-1	identify the advantages and disadvantages of store procedure and functions	08-09-2025	1							
10	37	4	1, 4	L-1	Subqueries: Comparator operators; subqueries: Single Row Subqueries	08-09-2025	1	As per Planned	Performing different operations on different databases	Solving different queries on different Db's functions,group discussion, activities	Lecturing, Demonstration, Activity, Interactive and Participative approach	Computers, Projector, Chalk & Talk, Youtube Videos	Students are familier with different functions of DB and SQL commands and queries	
	38			L-2	Multiple Row Subqueries	08-09-2025	1							
	39			L-3	Correlated subqueries; EXISTS, NOT EXISTS, ANY, ALL, SOME	08-09-2025	1							
	40			T-1	Presentation on the latest developments in research and industry related to this course	03-10-2025	1							

11	41	4	1, 4	L-1	Views: create, drop and update; realization of views based on single and multiple tables	09-09-2025	1	As per Planned	Lecture, Provide an exposure regarding Different platforms to collect information	Solving different queries on different DBS functions,group discussion, activities and class test	Lecturing, Demonstration, Activity, Interactive and Participative approach	Computers, Projector, Chalk & Talk, Youtube Videos	Students are familier with different functions of DB and SQL commands , queries and transaction operations	
	42			L-2	DCL: Controlling user access: Differentiating	09-09-2025	1							
	43			L-3	System privileges from object privileges; Granting privileges on tables	09-09-2025	1							
	44			T-1	Learn and present need of scalar subqueries	03-10-2025	1							
12	45	4	1, 4	L-1	PL/SQL: variables, datatypes; control statements (decision making)	15-09-2025	1	As per Planned	Lecture, Provide an exposure regarding Different platforms to collect information	Solving different queries on different DBS functions,group discussion, activities and class test	Lecturing, Demonstration, Activity, Interactive and Participative approach	Computers, Projector, Chalk & Talk, Youtube Videos	Students are familier with different functions of DB and SQL commands , queries and transaction operations	
	46			L-2	Stored procedures and Functions Concept; syntax and structure of store procedure	15-09-2025	1							
	47			L-3	syntax and structure of functions; calling a function; Examples	15-09-2025	1							
	48			T-1	Learn iterative statements in PL/SQL	06-10-2025	1							

13	49	4	1, 4	L-1	Managing and controlling transactions: Introduction of transaction, ACID properties;	16-09-2025	1	As per Planned	Lecture, Provide an exposure regarding Different platforms to collect information and perform transaction operations	Solving different queries on different DBS functions,group discussion, activities and class test	Lecturing, Demonstration, Activity, Interactive and Participative approach	Computers, Projector, Chalk & Talk, Youtube Videos	Students are familier with different functions of DB and SQL commands , queries and transaction operations	
	50			L-2	States of transaction; Transaction control; Overview of transaction management, using transaction control commands	16-09-2025	1							
	51			L-3	COMMIT, ROLLBACK, SAVE POINT, SET TRANSACTION; sufficient examples	16-09-2025	1							
	52			T-1	Does Relational model support storage of unstructured data, if no, what are the alternatives to store unstructured data.	06-10-2025	1							

Course Co-coordinator **Programme Co-coordinator** **Principal**